Unit 3, Lesson 6: Solving Two-Step, Real-World Problems with Fractions or Decimals – Part 1

Benchmark

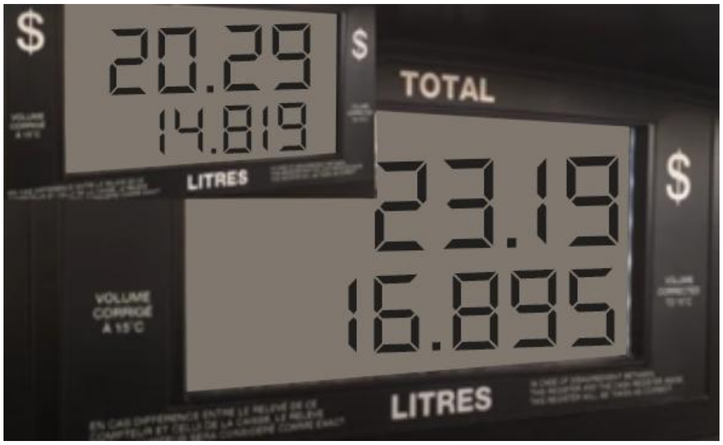
MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Learning Targets

- ☐ I can solve two-step, real-world problems involving any of the four operations with decimals.
- ☐ I can solve two-step, real-world problems involving any of the four operations with fractions.

3.6.1 Warm-Up: Gas Guzzler

The image below shows the cost of 14.819 and 16.895 liters of gasoline.



1. What is the price per liter of gas?

2. How much will it cost to pump 38 liters of fuel into the tank?

3.6.2 Guided Practice: Two-Step Rational Problems

1. Sophie grew her hair out for a long period of time. Her hair is $2\frac{3}{5}$ feet long. She donated her hair by cutting $\frac{5}{6}$ of a foot off. Sophie's hair then grew $1\frac{2}{3}$ feet.

    a. What operations are needed to determine the length of Sophie's hair?

    b. What values are necessary to determine the length of Sophie's hair?

    c. Write the numeric expression that can be used to determine the length of Sophie's hair.

    d. Determine the length of Sophie's hair.

    e. Determine if the length of Sophie's hair makes sense. Support your answer.

2. Tiffany has a balance of $\$167.32$ in her checking account. She deposits $\$58.16$ in her checking account and then writes a check for $\$97.27$.

    a. What operations are needed to determine the amount of money in Tiffany's bank account?

b. What values are necessary to determine the amount of money in Tiffany’s bank account?

c. Write the numeric expression that can be used to determine the amount of money in Tiffany’s bank account.

d. Determine the amount of money Tiffany has left in her checking account.

3.6.3 Collaboration: Problem Solving

1. A roll of ribbon was 12 meters long. Diego cut 9 pieces of ribbon that were 0.4 meters each to tie some presents. He then used the remaining ribbon to make some wreaths. Each wreath requires 0.6 meters.

    a. Draw a representation of the information.

    b. How does your drawing compare to your partner's drawing?

    c. Draw a circle around the piece of ribbon in your drawing that is available to make wreaths.

    d. Write the numeric expression that can be used to determine the amount of ribbon, in meters, available for making wreaths.

    e. How many meters of ribbon are available for making wreaths?

    f. What piece of information is not needed to determine the amount of ribbon available for making wreaths?

    g. Write a question that uses this information.

3.6.4 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

2. I can solve real-world problems with two steps using fractions.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.